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CO2- Assessment Tool-3

Name: Udhaya Kumar

Reg no: 192424332

Course Code: ITA408 Ethical Hacking

Topic 1: Foot Printing Tools - Conducting Competitive Intelligence...

A) Foot printing is the Process of Collecting publicly available Information About An Organizations without Direct interacting with its Systems. It helps Security Professionals understand an Organization External Exposure...

1.) Domain Registration Details:

Tools: WHOIS

Command

whois abc technologies.com

Information Obtained:-

- Domain Name
- Registrar
- Registration Date
- Expiry Date
- Name Servers

2) Employee Email Address:

the harvester - d o b c technologies . com - 6 all.

Information Obtained:

- Employee Email Address, • Hosts, • IP Address, public email IDs

3) Technology Stack:

Tools: • Netcraft, • Shodan.

Information Obtained:

- Web servers, • operating System, • Hosting Providers, • SSL certificate, • CMS.

Purpose: Identifies technologies used by Organization.

4) Publicly Exposed Subdomains

Tools: DNSDumpster.

Information Obtained:

- mail. obc technologies . com
- vpn . obc technologies . com
- post1 . obc technologies . com

Competitive Intelligence:

Tools

WITols

the Harvester

Netcraft

Shodan

DNSDumpster

Purpose:

Domain information

Email enumeration

Technology identification

Internet-Facing Services

Subdomain discovery.

Conclusion:

Passive Footprinting helps the identify available information allow organizations to Improve security information Exposure.

2) DNS Zone Transfer:

- A) A DNS Zone Transfer (AXFR) copies DNS records from the Primary DNS Server to a Secondary Server. Zone transfer can expose DNS Information:

Command:

```
dig NS example.com
```

Attempt Zone Transfer:

```
dig AXFR example.com @ns1.example.com
```

Information Identity:

- Internal Hosts
- Mail Servers
- Public Subdomains
- DNS Servers

Security Risks:

- Reveals internal network structure
- Helps attackers identify mail server and subdomains
- Can support phishing & impersonation information is publicly exposed.

Countermeasures:

- Disable unauthorized Zone Transfer
- Allow AXFR only for authorized Secondary DNS Server
- Enable DNSSEC
- Monitor DNS Logs, • Use Firewall Rules

Conclusion:

DNS Zone transfers should be restricted to authorized Secondary Servers to prevent unnecessary disclosure of DNS information.

3) Social Engineering Attacks and Countermeasures

A) The emails is a Phishing Attacks Because it impersonates the IT Departments and attempts to trick Employees.

Countermeasures:

- 1.) Verify the Sender's email address
- 2.) Do not click Unknown or suspicious links.
- 3.) Contact the IT Department through official Channels
- 4.) Check the Websites URL before Logging in.
- 5.) Never Share Passwords through email
- 6.) Report Suspicious For Emails immediately
- 7.) Enable Multi - Factor Authentication (MFA)

How These Action Prevent Credential Theft.

Countermeasure	Benefit
Verify Sender	Detect Fake emails
Avoid unknown Links	Prevent Fake Login Pages
Confirm with IT	Verifies authenticity.
Check URL	Identifies phishing websites
Never Share passwords	Protects password.
Enable MFA	Add an extra Layer of Security..

Conclusion

Phishing Attacks attempt to steal usernames and passwords through fraudulent emails. Following safe email practices, verifying requests and enabling MFA significantly reduces the risk of credential theft.